Vipransh Ojha

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Education

Vellore Institute of Technology — Bhopal, India
B.Tech in Computer Science and Engineering (AI & ML Specialization) — Sep 2023 – May 2027 (Expected)

Technical Skills

Languages: Python, C++, TypeScript, JavaScript, Java, SQL, HTML5, CSS3
Web Dev & Cloud: React, Node.js, Express.js, Flask, FastAPI, AWS, Docker, Linux, PostgreSQL, MongoDB
AI/ML & Data Science: TensorFlow, PyTorch, Scikit-learn, OpenCV, NLP, Pandas, NumPy
Tools & Libraries: OpenAI APIs, Unity, PyOpenGL, Git, REST APIs

Experience

Software Development Intern — New Delhi, India
BISAG-N (MeitY, Government of India) — May 2025 – Jun 2025

- Deployed a full-stack location intelligence and recommendation system for 500+ users, achieving 85% recommendation accuracy.
- Formulated content-based and collaborative filtering algorithms using Python and Scikit-learn for precise data targeting.
- Architected a highly scalable Flask backend with PostgreSQL, designed to process over 100 queries per second under load.
- Crafted a responsive interface with interactive spatial mapping, improving overall user engagement by 25%.

Research Intern (SERB Sponsored Project) — Bhopal, India
VIT Bhopal University — May 2024 – Jun 2024

- Created 'MolSpectra', a molecular simulation and visualization tool for a federally funded research project (Grant No. CRG/2022/002761).
- Engineered a seamless interface integrating 5+ quantum chemistry packages to automate complex simulation workflows.
- Constructed the application GUI utilizing PyQt and GLEW to streamline the high-fidelity 3D rendering of molecular structures.

Projects

Dikastirio – AI Assistant & VR Environment for E-Courtrooms — Jan 2026 – Present

- Developed a Unity VR spatial visualization environment, cutting critical document retrieval time by 40%.
- Implemented a local Retrieval-Augmented Generation (RAG) pipeline using LangChain and Llama 3 for secure, offline document analysis.
- Integrated natural language voice and gesture tracking controls, improving overall user experience and reducing interaction steps by 30%.
- Optimized rendering algorithms to maintain a smooth 90 FPS for large datasets on standalone hardware.

Automated Forensic Identification System — Aug 2025 – Oct 2025

- Built an end-to-end computer vision system achieving 92% accuracy in matching complex radiograph records.
- Trained and fine-tuned a Siamese Neural Network on the DENTEX dataset for high-precision comparative image analysis.
- Designed a robust Flask web interface enabling users to upload, process, and analyze data efficiently.
- Accelerated backend throughput to process 1,000+ images per hour, reducing manual processing time by 90%.

Interactive 3D Simulator & Algorithmic Solver — Oct 2023 – Dec 2024

- Programmed a 3D interactive simulator using Pygame and PyOpenGL, implementing complex mathematics for 360-degree rotations.
- Devised a high-precision object picking system via Ray-AABB intersection for real-time UI manipulation.
- Incorporated the Kociemba Algorithm to calculate optimal state-solving paths, solving any configuration in 20 moves or fewer.
- Improved computational performance by caching pruning tables, enabling near-instantaneous algorithmic resolution.

Achievements & Activities

- Founding Member & Student Coordinator, Agentic Intelligence Club
- Finalist in the National Buildathon organized by NASSCOM and Gnani.ai (selected from 40,000+ registrations).
- Solved 150+ algorithmic problems on LeetCode; active in coding competitions and hackathons.